Michael Farquharson

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Education

McMaster University Hamilton, ON
BEng in Mechanical Engineering and Management *Sept. 2015 – Apr. 2021*

- GPA 3.6/4.0 (Summa Cum Laude); entrepreneurship-focused engineering and commerce curriculum.

McGill University Montreal, QC
MSc in Mechanical Engineering *Sept. 2021 – Aug. 2025*

- Research in teleoperation, contact-rich simulation, and real-time control in partnership with NRC-AMTC.

Experience

HARP Technical Lead / Robotics Software Specialist Dec. 2023 – Present
Haply Robotics *Montreal, QC*

- Led development of the HARP teleoperation platform, including a Rust-based concurrent control interface for haptic devices.
- Architected low-latency control and modular safety features to support robust, scalable deployments.
- Managed a cross-functional team and supported demos, customer feedback, and technical stakeholder reviews.

Graduate Student Researcher Sept. 2021 – Aug. 2025
McGill University *Montreal, QC*

- Built teleoperation simulations with contact physics for remote aerial vehicles.
- Developed ROS 2-based test platforms for low-latency control experiments.
- Designed and evaluated haptic control algorithms for human-machine interfaces.

Data Analyst Intern May 2019 – Oct. 2019
Bombardier *Toronto, ON*

- Automated data cleaning and analysis workflows with Python to drastically reduce recurring task time.
- Delivered quarterly presentations to senior management highlighting efficiency gains.

Projects

HARP Teleoperation Control Platform | *Rust, Concurrency, Haptics* 2025 – Present

- Designed a low-latency control platform for force-feedback teleoperation devices.

Teleoperation Research Platform | *ROS 2, Simulation, Haptics* 2022 – 2025

- Built contact-rich simulations and a ROS 2 testbed to study haptic teleoperation.

Custom Mechatronic Prototyping | *3D Printing, Control, Hardware* 2023 – 2024

- Rapidly prototyped interface devices and iterated on mechanical and control system designs.

Arduino Mechanical Clock | *CAD, 3D Printing, C++* 2020

- Built a mechanical clock with stepper motors and 3D-printed gear trains driven by Arduino.

Technical Skills

Languages: Rust, Python, C/C++, MATLAB, Bash

Robotics & Control: Teleoperation, Haptics, ROS 2, Control Systems, Real-time Simulation

Tools: Git, Linux, SolidWorks, Onshape, Autodesk Inventor

Prototyping: 3D Printing, Soldering, Rapid Prototyping, Mechatronic Integration